

From Text to Torque: Feasibility-Aware Tracking of Text-Generated Humanoid Motions

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Main finding

Text-generated humanoid motion is stochastic enough that **reference choice becomes an RL intervention**. Before running expensive PPO, cheap motion diagnostics can select generated seeds that are easier to track physically. **Diagnostic-best references beat diagnostic-worst references on jump, roll, and squat strict-PPO probes**.

Problem: plausible motion is not physical control

Text-to-motion models can generate convincing kinematic humanoid clips, but a torque-controlled Unitree G1 policy must satisfy contacts, gravity, balance, joint limits, and termination thresholds.

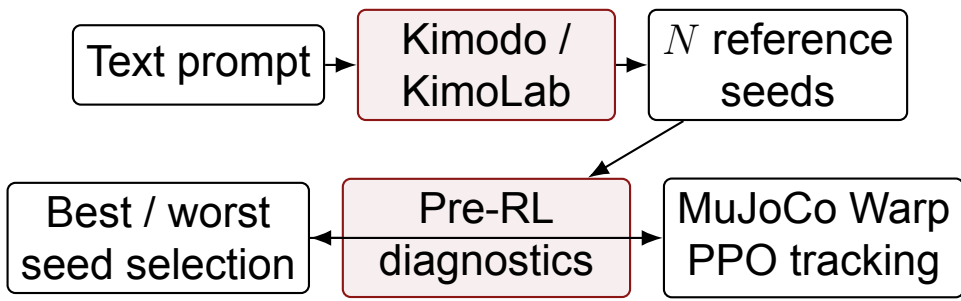
Research question: which properties of a text-generated reference motion predict whether PPO can learn a stable tracking policy? We treat the generator as a stochastic proposal distribution over references, not as a guaranteed controller.

Hypothesis: a reference feasibility boundary

PPO succeeds when the generated reference lies inside a reachable tracking basin. It fails when the kinematic target asks for states or contacts that the robot cannot realize under torque, balance, and termination constraints.

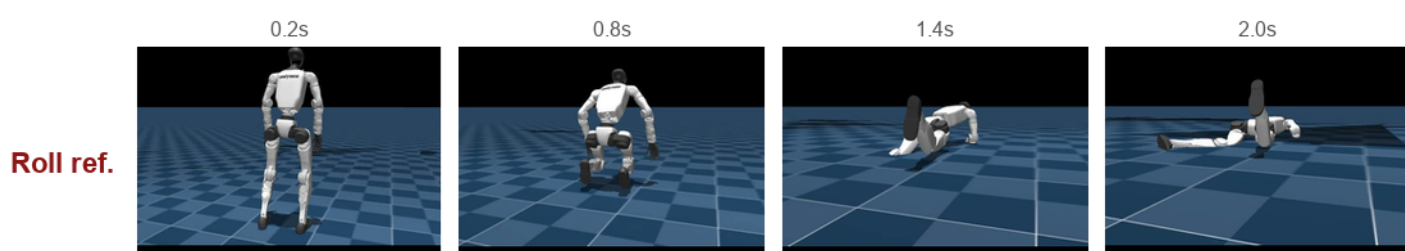
Reference signal	Expected PPO symptom
root / joint acceleration spikes	high action-rate cost, falls, poor velocity tracking
bad contact or body pose	self-collisions, early terminations, low body-pose reward
large seed-to-seed variation	best-of-N selection should improve learnability
mostly speed-limited failure	temporal retiming should help

System pipeline



We run KimoLab on Modal GPUs, convert generated CSV motions to MuJoCo reference trajectories, and train PPO policies to track them in simulation.

Reference motions before physics

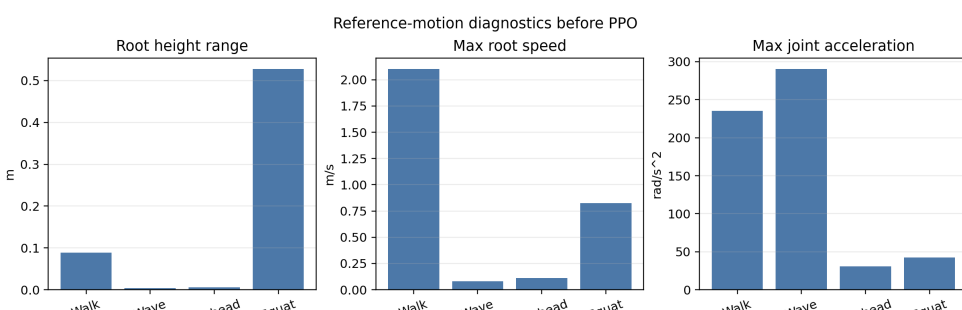


Generated references can look plausible while already containing balance, contact, or body-pose artifacts that downstream PPO must absorb.

Diagnostics before PPO

For each generated motion, we compute cheap reference-only features:

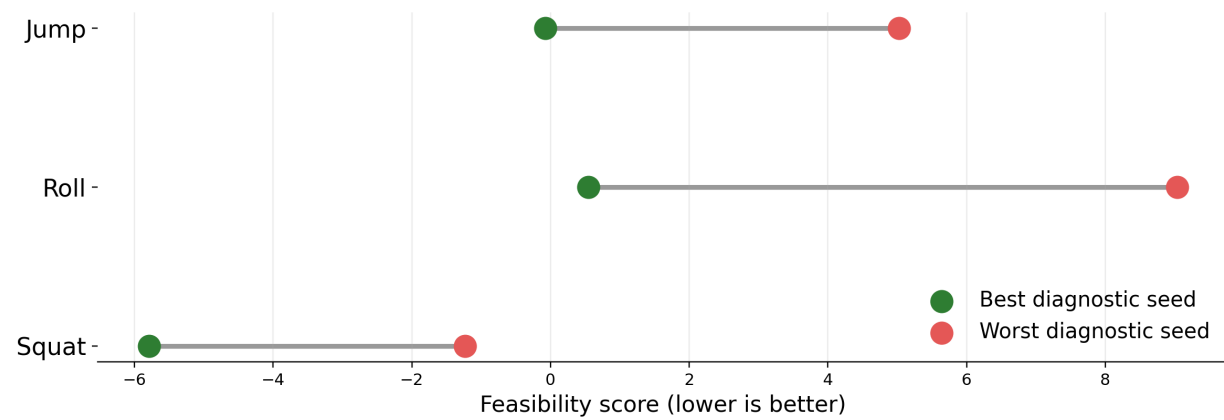
- root height range, root speed, root acceleration
- joint velocity and acceleration extrema
- body angular speed and initial-state feasibility checks



Diagnostics reveal that failure is not just "hard prompt" vs "easy prompt"; generated seeds vary materially within the same prompt.

Best-of-N reference selection

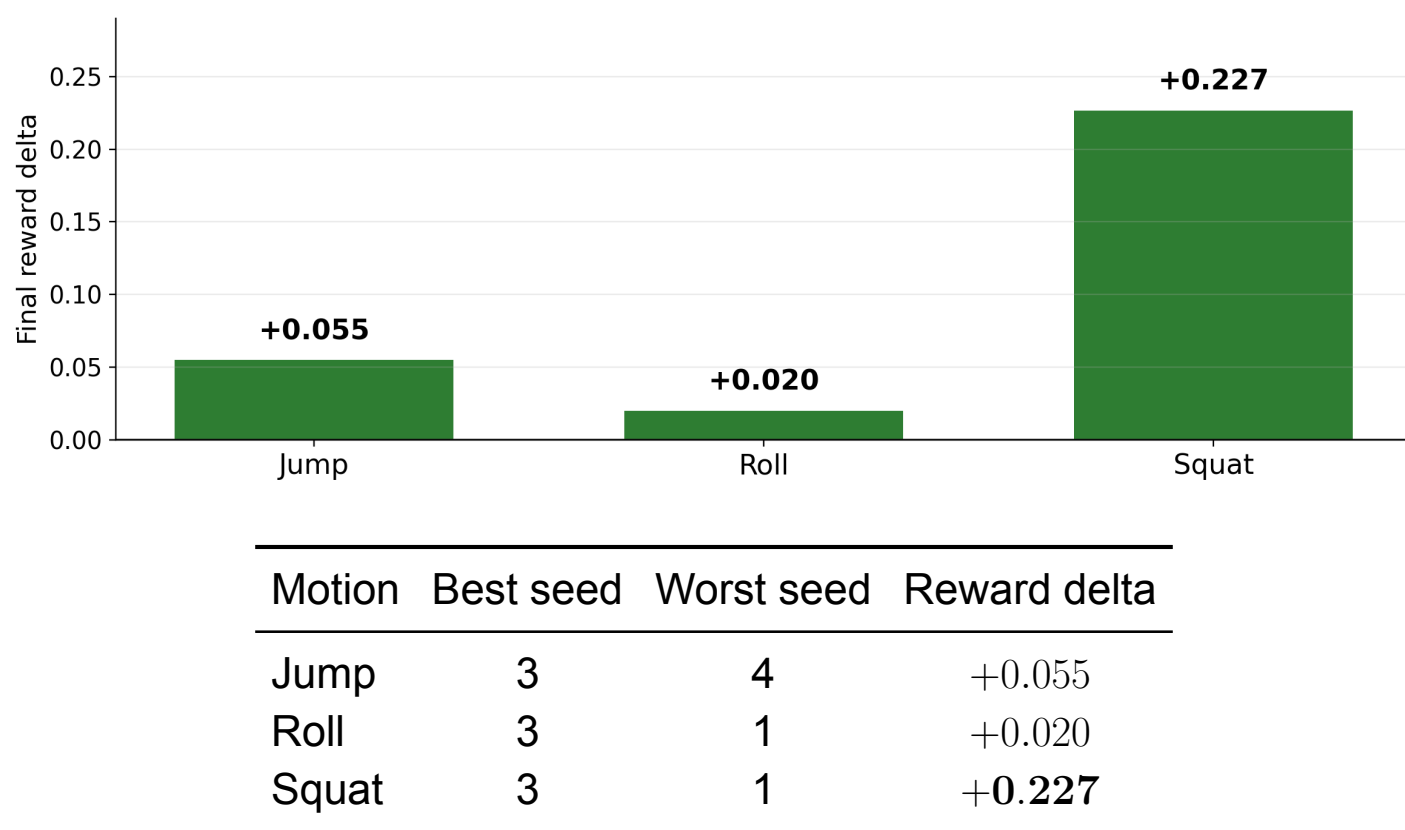
For each prompt, generate 3–5 motion seeds, score every reference before PPO, then compare strict-PPO probes for the diagnostic-best and diagnostic-worst seeds. This tests whether the generator's randomness can be exploited as a search space over feasible references.



Lower feasibility score is better. Same prompt, different generated seed, very different reference feasibility.

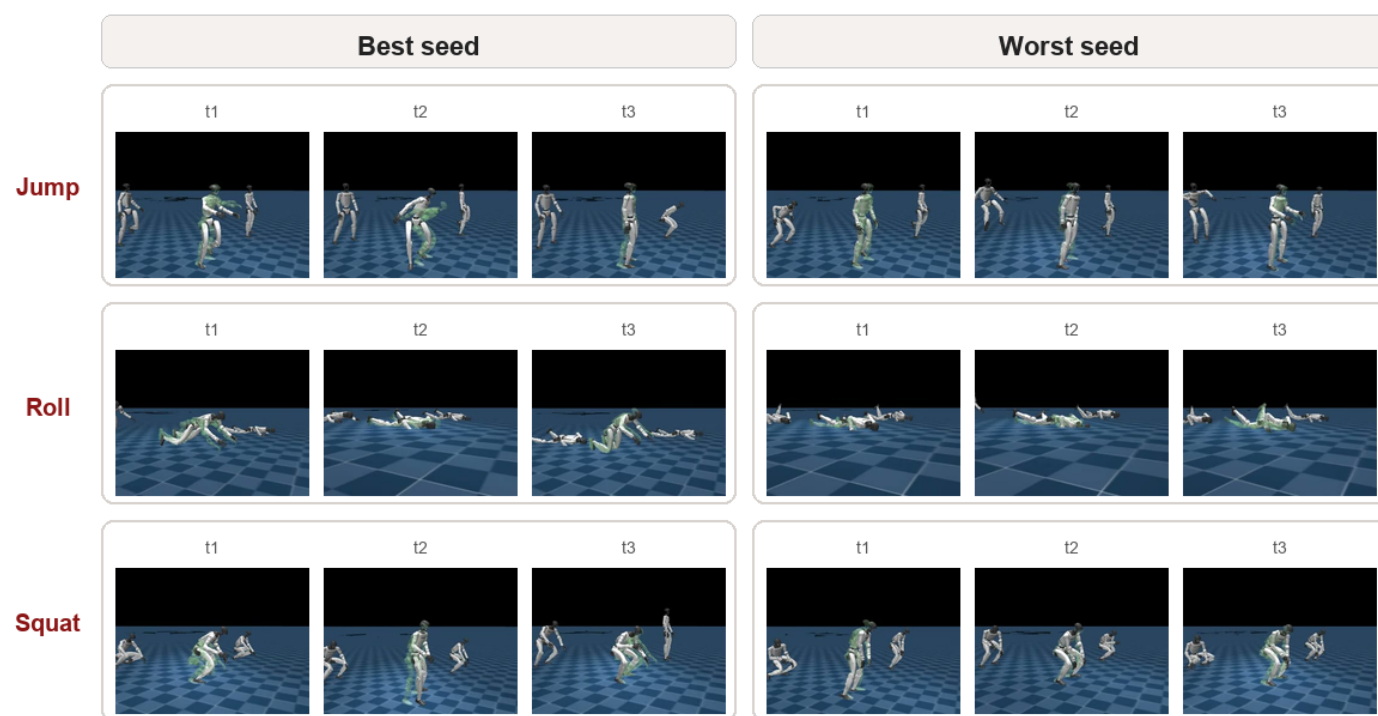
The feasibility score is a clipped robust-z sum over reference-only diagnostics such as root acceleration and max joint acceleration. The point is not to learn a black-box predictor; it is to cheaply reject bad references before RL.

Result: diagnostics predict early learnability



Diagnostic-best seeds improve strict-PPO reward on every tested motion. This supports the feasibility-boundary view: same prompt semantics, different generated dynamics, different PPO outcome. The practical recipe is simple: sample a few references, score them, and train the one most likely to be physically learnable.

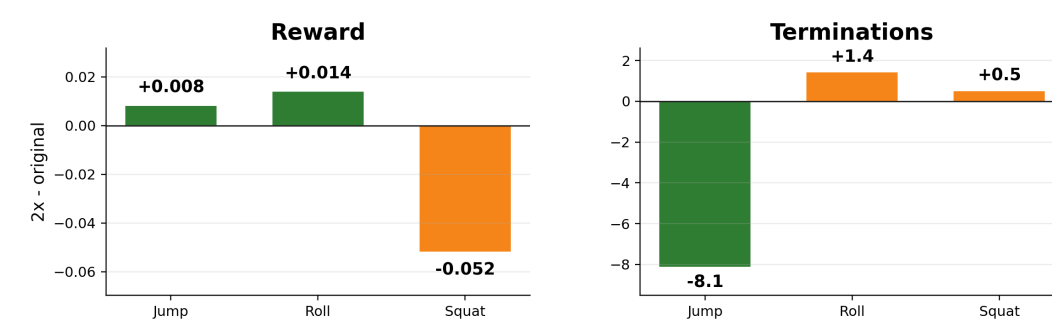
Qualitative evidence



Three frames per rollout at the same probe horizon. Best and worst generated references produce visibly different early tracking behavior.

Alternative repair: slow the reference

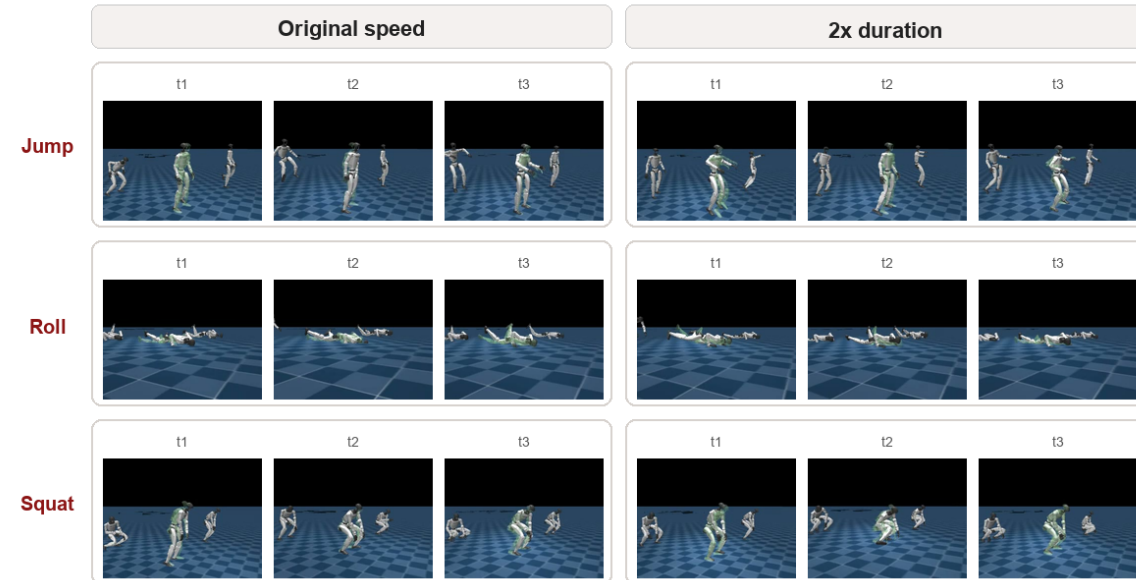
We also tested a temporal feasibility repair: take the diagnostic-worst seed and train strict PPO after slowing the reference to 2× duration. This asks whether failures are mainly caused by excessive speed or torque demand.



Motion	Reward delta	Termination delta	Verdict
Jump	+0.008	−8.125	slight help
Roll	+0.014	+1.417	mixed
Squat	−0.052	+0.500	worse

Naive retiming alone is not a reliable repair. It slightly helps jump and roll reward, but all slowed worst-seed runs still terminate early. This negative result matters: hard generated references are not only too fast; many also encode poor contacts or awkward body states.

Rollouts: temporal repair is mixed



Original-speed worst seeds vs 2× slowed worst seeds. Retiming changes behavior, but does not solve the hard references.

Diagnostic-gated training recipe

Diagnosis	Action before expensive PPO
Low feasibility score	run strict tracking baseline
Borderline score	compare strict vs loose/adaptive termination
High score or contact artifacts	regenerate seeds; avoid training on the worst sample
Large speed / acceleration spikes	test retiming as an ablation, not as the default fix

The pipeline becomes a feasibility-aware loop: generate seeds, score them, choose the intervention, then spend GPU time.

Experimental protocol

Suite	walk, wave, tap head, squat, jump, roll, acrobatic references
Probe	150 strict-PPO iterations, 256 parallel environments
Metrics	reward, tracking errors, episode length, terminations, videos

Takeaways

- Text-to-motion quality should be evaluated by **downstream physical learnability**, not only visual plausibility.
- Best-of-N reference selection is cheap, simple, and improved all tested strict-PPO probes.
- Naive temporal repair is mixed; bad generated references often need regeneration or adaptive training, not just slower playback.
- Main demo message: reference generation is part of the RL algorithm, not a preprocessing detail.